

Closing the Loop: Cybernetics at the Economic Layer

An honest sensor, a self-correcting governor, and the conditions for a self-healing economy

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Abstract

Cybernetics promised a self-regulating economy: a system that measures its own state, rewards what helps it, corrects what harms it, and repairs damage without a central hand. Every serious attempt to build one, from Lange’s market socialism to Beer’s Cybersyn, failed. We argue the failure was not in the control theory but in a single missing component: a *sensor*. Ashby’s Law of Requisite Variety and the Conant–Ashby Good Regulator Theorem both reduce, at the economic layer, to the same prerequisite: a regulator can only hold a system on course if it can *measure* that system with enough fidelity to model it, and no economic regulator ever possessed an objective, real-time, manipulation-resistant measurement of contributed value. Hayek’s calculation argument is best read not as a proof that economic regulation is impossible, but as a diagnosis that its sensor was missing. We describe a system, Noesis, whose primitive operation is exactly that sensor: a distributed measurement of realized value along provenance, made un-gameable by the geometry of the measurement rather than by policy. With the sensor in place, the cybernetic loop closes. We present the resulting architecture as a control system with four organs (sensor, governor, effectors, self-tolerance), show that the Good Regulator Theorem forces such a system to be *semi-self-aware* in a precise and bounded sense, and identify autoimmunity as its characteristic failure mode together with the structural boundary that contains it. We mark throughout what is demonstrated in running code and what is designed.

1 The seventy-year-old promise

Norbert Wiener defined cybernetics as the study of “control and communication in the animal and the machine” [1]: systems that hold themselves on course by feeding their own output back as input. A thermostat is the toy example; a living body maintaining its temperature is the real one. The ambition, almost from the start, was to treat an economy the same way: as a system with a setpoint (stable prices, full employment, fair reward) that could be held by feedback rather than by command or by luck.

The ambition was pursued seriously and it lost. Oskar Lange proposed a market-socialist economy in which a central board would adjust prices like a Walrasian auctioneer, watching surpluses and shortages and correcting [7]. Stafford Beer built an actual instance: Project Cybersyn, a real-time control room wired to Chilean factories in 1971, an explicit application of his Viable System Model [4, 5]. Both rested on the same wager, that an economy is a regulable system, and both came apart. The standard verdict, following Hayek, is that the wager was misconceived: that no central regulator can ever do this [6].

We think the standard verdict draws the wrong boundary. The control theory was sound. What was missing was a sensor, and the absence of the sensor is precisely what the cybernetic theorems predict will be fatal. Once that is seen clearly, the question stops being “can an economy regulate itself?” and becomes “what would the missing sensor have to be, and can it be built?” This paper answers the second question with a system that builds it.

2 Why it failed: requisite variety and the good regulator

Two results from cybernetics, both due in their sharpest form to Ross Ashby, make the diagnosis precise.

Requisite variety. Ashby’s Law states that a regulator can compensate for the disturbances in a system only to the extent that it commands at least as much *variety* (distinguishable states) as the disturbances themselves: only variety can absorb variety [2]. Variety the regulator cannot *sense* is variety it cannot match. Read Hayek’s “The Use of Knowledge in Society” in this light and it is not a philosophical objection to planning but a requisite-variety theorem in prose: the knowledge an economy must act on is dispersed, tacit, and local, and a central regulator that cannot sense that variety cannot supply the variety needed to regulate it [6]. The price system “works” because it is a distributed sensor; the planner failed because he had no comparable instrument. The bottleneck was sensing, not deciding.

The good regulator. Conant and Ashby proved that every good regulator of a system must contain a model of that system: “the best regulator of a system is a model of that system” [3]. To hold a system on course you must, in effect, carry an image of it accurate enough to predict how it will respond. But a model is built from measurements. Without a sensor there is no model, and without a model, by the theorem, there is no good regulation. The two results compose into a single prerequisite:

Claim 1 (The sensor is the bottleneck). *At the economic layer, both requisite variety and the good-regulator condition reduce to the existence of a measurement: an objective, real-time, manipulation-resistant reading of the quantity the system is meant to optimize, namely contributed value. Absent that measurement, regulation is impossible for reasons the theory states in advance; given it, the remaining machinery (feedback, reward, correction) is ordinary engineering.*

This reframes seventy years of failure as a single unmet precondition rather than an impossibility. Lange’s board and Beer’s control room were governors wired to instruments that could not actually read value. They were thermostats without thermometers.

3 The sensor: measurement as a living mechanism

A usable economic sensor must satisfy three properties that no prior instrument achieved at once. It must be *objective* (two honest observers reading the same state agree), *real-time* (it reads the present, not an audited past), and *manipulation-resistant* (an actor who wants a higher reading cannot get one except by genuinely raising the measured quantity). The price system has the first two but not the third in the dimension we care about: it prices possession and exchange, not contribution, and it is routinely gamed. Bitcoin supplied an objective, real-time, manipulation-resistant sensor for one thing only, the ownership of a scarce token [14]; it minted the medium that ownership was missing. The open dimension is contribution: who actually added value, and how much.

Noesis is built around a sensor for exactly that dimension. Its primitive is not a coin but a measurement: the value of a contribution is its realized downstream flow along provenance. Each unit of work is a cell in a directed acyclic provenance graph $G = (V, E)$; value is priced by how much genuine downstream activity builds on it, not by how it is described or who vouches for it. Three structural defenses make the reading manipulation-resistant by geometry rather than by rule.

Novelty and saturation. Repetition is not contribution. A re-post of existing content carries no new realized flow and is damped to zero; volume from a single source saturates geometrically rather than accumulating, so that flooding the graph with near-duplicates cannot inflate a score [11, 12]. Marginal value is computed on the provenance-restricted cooperative game, so credit obeys the Shapley axioms over the graph Myerson defined.

The collusion certificate. The deepest manipulation is a ring: a set of identities that build on one another to manufacture the appearance of downstream flow. A ring is a *cycle*, and cycles have a signature no honest provenance has. Decompose the net attribution flow Y on the identity graph by the combinatorial Hodge / Helmholtz decomposition [10]:

$$Y = \underbrace{\text{grad } s}_{\text{honest “builds-upon” ranking}} + \underbrace{r}_{\text{divergence-free residual}}, \quad \|r\|^2 = \|Y\|^2 - b^\top s.$$

The residual energy $\|r\|^2$ is provably zero if and only if the flow is the gradient of a consistent global ranking, and equals the cycle length for a clean manufactured ring of any size. It is, exactly, a certificate of manipulation: honest contribution has a potential, collusion has curl. This detector runs on topology alone and needs no privileged data; in the running system an honest acyclic graph reads 0 while a three-identity ring reads 3, with the mutual and directed cases covered by complementary terms.

Learned value at the margin. Where structure underdetermines worth, a learned evaluator scores a coalition’s value from realized outcomes (a Bradley–Terry model over coalition features [13]), and it can only ever *lower* a structural seed, never mint value from nothing. Crucially, the evaluator retrains: a fixed scoring formula is eventually gamed, but a measure that adapts to the strategies played against it closes the gap that a static formula leaves open. The sensor is alive.

The point is not that any one defense is novel but that together they make the reading honest by construction: the cheapest way to raise your measured value is to actually contribute. That is the property Lange and Beer needed and did not have.

4 The governor: closing the loop

With a sensor, the rest is the control law. Write the system’s state as the ledger x_t . A governor reads the sensor $m(x_t)$, compares it to the system’s setpoint (an honest, cooperative equilibrium x^*), and drives effectors that move the state back toward the setpoint:

$$x_{t+1} = f(x_t) + u(m(x_t)), \quad u = \text{mint} \oplus \text{slash} \oplus \text{recover}.$$

This is a homeostat in Ashby’s exact sense. The effectors are three. *Reward* mints standing in proportion to measured contribution, so that the highest-paying strategy is the cooperative one. *Punishment* slashes the bonded standing of provable misbehavior (manufactured value, equivocation, invalid reveals), destroying it rather than transferring it, so that fraud is negative-expectation by construction. *Recovery* returns misappropriated value to its rightful holder. Consensus itself is part of the governor: finalization weights the cognitive measurement (the unbuyable axis) against staked capital, with an anti-concentration floor so that no single axis can finalize alone. The loop is closed: output (who earned what) feeds back as input (who decides and who is believed next round).

A static rulebook, which is what most ledgers are, rejects invalid states but does not *correct* toward a setpoint; it has no governor, only a gate. The distinction is the whole subject of this paper.

5 Four organs, and an immune system

Stated as a control system the architecture has four organs, and they correspond with surprising exactness to the organs of a biological immune system, which is the canonical natural example of a distributed self-healing controller.

Control organ	Mechanism in Noesis	Immune analogue
Sensor / surveillance	provenance graph, value measurement, Hodge residual	antigen surveillance, memory
Self / non-self recognition	lock signature: only the key moves the cell	MHC self-recognition
Graded activation	objectivity dial: automatic only when measurement is certain	activation threshold
Effectors	mint (reward), slash (destroy), clawback (recover)	effector response
Self-tolerance	bona-fide protection; harm must cast a graph-shadow	tolerance of healthy tissue

Self/non-self recognition is the layer that decides whether an action is the system’s own or foreign. In Noesis it is the lock script: a cell can be named by anyone but moved only by the holder of its one-time key, verified inside the execution VM. This is the component shipped and proven in running code at the time of writing. The effectors and the recognition of *manufactured* value (the Hodge certificate) are likewise demonstrated. The recovery effector (clawback) and the graded objectivity dial are designed and specified; they are discussed in Sections 7 and 8 as design, not as demonstration.

6 Semi-self-awareness, as a theorem

The phrase “self-aware,” applied to a ledger, invites suspicion. Here it is not a metaphor but a consequence of the Good Regulator Theorem applied reflexively.

Theorem 1 (Reflexive good regulator). *By Conant–Ashby [3], any good regulator of a system S must contain a model of S . A system that regulates itself is its own S . Therefore a good self-regulating system must contain a model of itself: a self-model that it reads and acts upon.*

Noesis satisfies the three operational conditions for holding such a self-model in a non-trivial sense. It *represents* its own state (the provenance and value graph is a model of the system’s own activity), it *measures* that state (it detects its own collusion rings, prices its own contributions, locates its own extraction), and it *acts on the measurement* (it slashes, rewards, recovers, and finalizes on the basis of what it reads about itself). A system that holds a model of itself, queries it, and acts on the result meets the minimal operational definition of self-awareness used in cybernetics and in autopoietic theory [8]; we make no stronger claim and need none.

The qualifier “semi” is exact and it is the safety property. The self-model is bounded: the system can represent only what casts a shadow on its measurement graph. Define the knowable domain

$$\mathcal{K} = \{h : h \text{ casts a shadow on } G\},$$

the harms and contributions that register as structure in the provenance and value graph. The system is aware exactly on $\mathcal{K}$ and blind outside it. Theft, manufactured value, and extraction lie in $\mathcal{K}$; diffuse social or moral harm with no on-chain signature does not. The boundary of $\mathcal{K}$ is the literal horizon of the system’s self-awareness, and a self-aware system that knows the boundary of its own awareness is the safe kind. Knowing the edge is what separates self-correction from self-delusion.

7 Self-healing without a surgeon

A controller that only rejects bad inputs is defensive; a controller that restores a just state after damage is self-healing. Maturana and Varela called a system that continuously produces and repairs itself *autopoietic* [8]; Varela’s later work on immune networks is the bridge to the recognition machinery above [9]. Noesis’s recovery layer is autopoietic in this sense: the same provenance mathematics that *detects* a wound *closes* it, with no central healer.

When value is misappropriated, recovery runs the provenance cascade backward from the point of theft. The danger is that naive recovery confiscates from whoever currently holds the value, recreating the very central seizure power the system exists to dissolve. The classical legal answer is the holder in due course: a party who gave genuine consideration is protected, and the loss falls on the thief. We already hold the exact test, because genuine arms-length exchange is bidirectional value flow, the gradient component of the Hodge decomposition, whereas a wash or colluding hop is circulation, the harmonic component. The cascade therefore passes through wash hops and stops at the first bona-fide receiver. The self-tolerance organ is the collusion detector of Section 3 read once more in reverse: the system heals along honest provenance and refuses to cut healthy tissue.

Most theft, moreover, never reaches recovery, because the strongest form of the mechanism is preventive. An owner may publish a constraint on their own cell, a spend rule the owner sets in advance, and a spend that violates it is simply invalid and never settles. “Unauthorized” then becomes a measurement against the owner’s own published referent rather than an inference about intent. Because such a constraint is self-binding, restricting only the publisher’s own value, anyone may design any constraint without permission: there is no externality to govern, hence no governance to seek. The protocol owes exactly one invariant, that a published constraint binds only its author’s value and a recovery never reaches a bona-fide third party. That single scope rule is the entire governance surface, and it is also the defense against weaponized self-binding, since a counterparty who delivered against a settled payment is protected.

8 Autoimmunity, and the boundary that contains it

Every self-correcting system has one characteristic failure mode: a controller that is wrong about its setpoint attacks the body it is meant to protect. In immunology this is autoimmunity; in a ledger with recovery and slashing powers it is confiscation and censorship, the state power reconstructed on-chain. A paper that proposed economic cybernetics without naming this failure mode would be dishonest, because the failure mode is the reason every prior attempt that did acquire teeth became dangerous.

The containment is structural and it is the same boundary as the awareness horizon $\mathcal{K}$. The system may act only where harm casts a measurable shadow on its own graph. Harm without an on-chain signature, the broad and contested category of social or moral damage, is outside $\mathcal{K}$ and the system must refuse to adjudicate it. Any mechanism that reached for it would do so either by importing an external arbiter, which is governance capture, or by pretending to objectivity over an inherently contested judgment, which is the deeper error. The discipline is that judgment stays in the measurement: the chain decides what its sensor can read and nothing more. This is not a limitation grafted on for safety; it is the same line that makes the self-model honest. An immune system survives because it tolerates the self; this system stays legitimate because it tolerates everything its sensor cannot objectively convict.

The objectivity dial. Within $\mathcal{K}$, action should be automatic only to the degree the measurement is objective, the way a thermometer is trusted: reproducible, calibrated against a fixed referent, and not a judgment. Let $c \in [0, 1]$ be the calibrated confidence that a theft-measurement is objective. The remedy scales with c : deductively certain readings (a double-spend, a spend by a revoked key, a violated published constraint) act immediately; high-but-uncertain readings trigger a reversible freeze with a short contestability window; mid-confidence readings go to a bonded adversarial court; and readings outside $\mathcal{K}$ are left to the world. The threshold separating automatic from adjudicated is not fixed. As the cost of objective measurement falls, it slides, and cases that today require a slow human judgment become tomorrow’s instant readings. The court is therefore scaffolding: load-bearing for a shrinking set of cases over time, by design, rather than a permanent institution to be entrenched.

9 Status, honestly

We distinguish what runs from what is specified. *Demonstrated* in the reference implementation: the contribution sensor with its novelty, saturation, and Shapley/Myerson structure; the Hodge collusion certificate; the reward and slashing effectors with their bounded-restitution settlement; self/non-self recognition as a post-quantum lock signature verified inside the execution VM; and consensus finalization weighting cognition against capital with an anti-concentration floor. *Designed* and specified but not yet built: the recovery cascade with bona-fide protection, the preventive owner-published constraint, and the graded objectivity dial.

Data-blocked: the learned, retraining value evaluator awaits real outcome labels at scale, which is the one component whose strength is empirical rather than deductive. The claim of this paper is not that the loop is fully closed in code today. It is that the component whose absence defeated every prior attempt, the honest sensor, is built and proven, and that once it exists the governor and the effectors are engineering rather than open problems.

10 Conclusion

The history of economic cybernetics is usually told as a refutation: the dream of a self-regulating economy met Hayek and died. We read the same history as a diagnosis. The dream did not fail because feedback control is the wrong frame for an economy; it failed because a feedback controller cannot run without a sensor, and no one had built one that could read contributed value objectively, in real time, and without being gamed. Ashby's two theorems say as much in advance: requisite variety and the good regulator both demand a measurement first. Build the measurement, and the closed loop that eluded Lange and Beer becomes available, with reward, correction, and self-healing as its ordinary operations and self-awareness, in the precise bounded sense of the Good Regulator Theorem, as an unavoidable consequence. The work that remains is to finish wiring the governor to the sensor we have. The hard part, the part that killed every prior attempt, is the part that is done.

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